

Research & Strategy Report

A Premium Local Dashboard for Ideogram 4

Smoothing the JSON-prompting & bounding-box pain — and driving hyper-realistic generation entirely on a local RTX 4090, no cloud.

THE THESIS

Ideogram 4's biggest frustrations — mandatory JSON, fiddly hand-typed bounding boxes, whole-image regen on every edit, expiring credits, queues, and a censorship gate — are **mostly UX and delivery problems, not model problems**. Ideogram 4 shipped as **open weights** that run on a 24 GB 4090. So we can keep the model's strengths (best-in-class text + native layout control) and wrap them in a **visual, local, no-credits dashboard** that fixes the delivery layer. This report covers the top 10 pain points, the local engine feasibility, a realism pipeline, and a build plan.

READ FIRST — LICENSING

The public Ideogram 4 weights are released under a **non-commercial** model agreement. Personal/research/local use is fine; any client-facing, internal-business, or revenue-linked use needs Ideogram's **paid commercial license** (which also unlocks full-precision weights). Verify `LICENSE.md` on the HuggingFace repo before building anything commercial on these weights.

Prepared June 2026. Findings from Reddit/X, GitHub, HuggingFace, ComfyUI & Civitai research; sources on the final pages. Post-release figures are directional — flagged in text.

1 · Executive summary

- **The opportunity.** No existing local tool offers a clean **visual bbox** → **structured-JSON** editor. Ideogram 4 is the first model where that editor is genuinely valuable — and it's now self-hostable. That's the gap to own.
- **Feasibility: yes, on a 4090.** Ideogram 4 is a 9.3B diffusion transformer. A 4-bit build (NF4, or GGUF Q4_K for better text) fits 24 GB when the Qwen3-VL text encoder is CPU-offloaded. Expect ~20–70 s at 1K, ~60 s at 2K.
- **Engine: ComfyUI as a local backend** (day-0 Ideogram-4 support) behind a custom premium UI — don't reinvent diffusion; reinvent the experience.
- **Realism path (single-model).** No second engine. Realism comes from a **native Ideogram-4 realism LoRA** (the only kind that loads), **photo-style JSON prompt craft**, and IG-4's own refine/inpaint passes — plus a non-generative upscaler. Popular Flux/SDXL Civitai LoRAs can't load on IG-4 and are excluded.
- **What local fixes:** credits, queues, privacy, the censorship gate, and — via the visual editor — the JSON/bbox pain. What it can't fix: the non-commercial license and the model's photoreal ceiling.

At-a-glance verdict

Question	Verdict
Run Ideogram 4 locally on a 4090?	Yes — 4-bit only; offload the text encoder; batch size 1 at 2K.
Best checkpoint?	NF4 for compatibility; GGUF Q4_K for better text fidelity at the same VRAM.
Fix JSON/bbox pain?	Yes — a visual canvas that emits correct [y, x, y, x] JSON.
Hyper-realism?	Partly — IG-4-native realism LoRA + photo-style JSON + IG-4 refine/inpaint. No second model.
Ship commercially on open weights?	No — non-commercial license; needs paid tier.

2 · The top 10 Ideogram 4 pain points (Reddit / X / forums)

Ranked by how often and how loudly users raise them, with the segment each hurts most.

#	Pain point	What users say	Hurts
1	JSON prompting feels mandatory & brittle	Docs admit plain text underperforms and trips the safety filter more — users feel forced into authoring/maintaining JSON 'design briefs.'	Hobbyists
2	Bounding-box placement is fiddly	Coordinates are hand-tuned on a 0–1000 grid; overlapping boxes melt typography. No live visual feedback in the base product — guess, generate, inspect, repeat.	Designers
3	Coordinate order confusion	bbox is y-first [y_min, x_min, y_max, x_max] — reverse of the x-first convention; swaps silently misplace elements.	Developers
4	JSON edits regenerate everything	Changing one field re-runs full inference; the whole composition shifts. Localized fixes need a separate Magic-Fill/inpaint detour.	Designers
5	Expiring credits & surprise pricing	Free tier repeatedly cut; priority credits don't roll over; some changes shipped unannounced (incl. to paid users).	Hobbyists
6	Slow / unpredictable queue	Slow-queue waits ballooned (reports up to ~20 min); throughput as low as a few images/hour at peak.	Marketers
7	Aggressive safety filter	Benign prompts blocked inconsistently (documented GitHub examples); false-positive rate higher for non-JSON prompts.	All
8	'Open-weight' is non-commercial	Weights are downloadable but non-commercial; widely mis-read as open source. Commercial use needs the paid tier.	Developers
9	Photoreal / consistency / weak seeds	Strong on type, weaker on photoreal & character consistency; seeds give poor reproducibility.	Designers
10	Flat output — no layers	Single flat PNG; logos arrive with artifacts, no vector/layers — assets get rebuilt in a real editor.	Designers

WHAT USERS WISH EXISTED

A drag-and-drop bbox canvas that writes the JSON for them · true localized edits (change one element, freeze the rest) · no credits/queues/surprise pricing · transparent, tunable safety · layered/editable output.

A local dashboard delivers four of these five.

3 · What a local dashboard actually fixes

Honest mapping of each pain point to whether self-hosting on your own GPU solves it.

Pain point	Local fix?	Why
JSON mandatory/brittle	Partly	UI auto-builds & lints JSON from plain text; model's JSON reliance remains.
Bbox fiddliness	Yes	Drag-to-draw canvas emits coordinates — pure UX layer, fully solvable.
Coordinate-order confusion	Yes	Visual editor never exposes raw [y,x,y,x]; the app writes it correctly.
Edit regenerates all	Partly	Wire in mask/inpaint + seed-lock for region edits; not a true layered edit.
Expiring credits / pricing	Yes	Own-GPU inference removes credits, billing, and surprise changes.
Slow queue / speed	Yes	No shared queue; bounded only by your hardware.
Safety-filter false positives	Partly	Self-hosted weights skip the hosted gate; you own compliance + license risk.
Non-commercial license	No	A dashboard can't change the license; commercial use needs the paid tier.
Photoreal / consistency / seed	Partly	True deterministic seeds + easy A/B; photoreal ceiling is model-bound.
Flat / no layers	Partly	Chain bg-removal / segmentation / vectorize; native layers await next model.

Scorecard: **4 fully solved** (bbox, coordinate order, credits, queue), **5 smoothed**, **1 out of scope** (license). The fully-solved set is exactly the set people complain about most — which is what makes this product compelling.

4 · Proposed system architecture

A thin, premium native shell over a local ComfyUI engine. The UI owns the experience; ComfyUI owns the compute. Everything runs on the one machine — no network calls leave it.

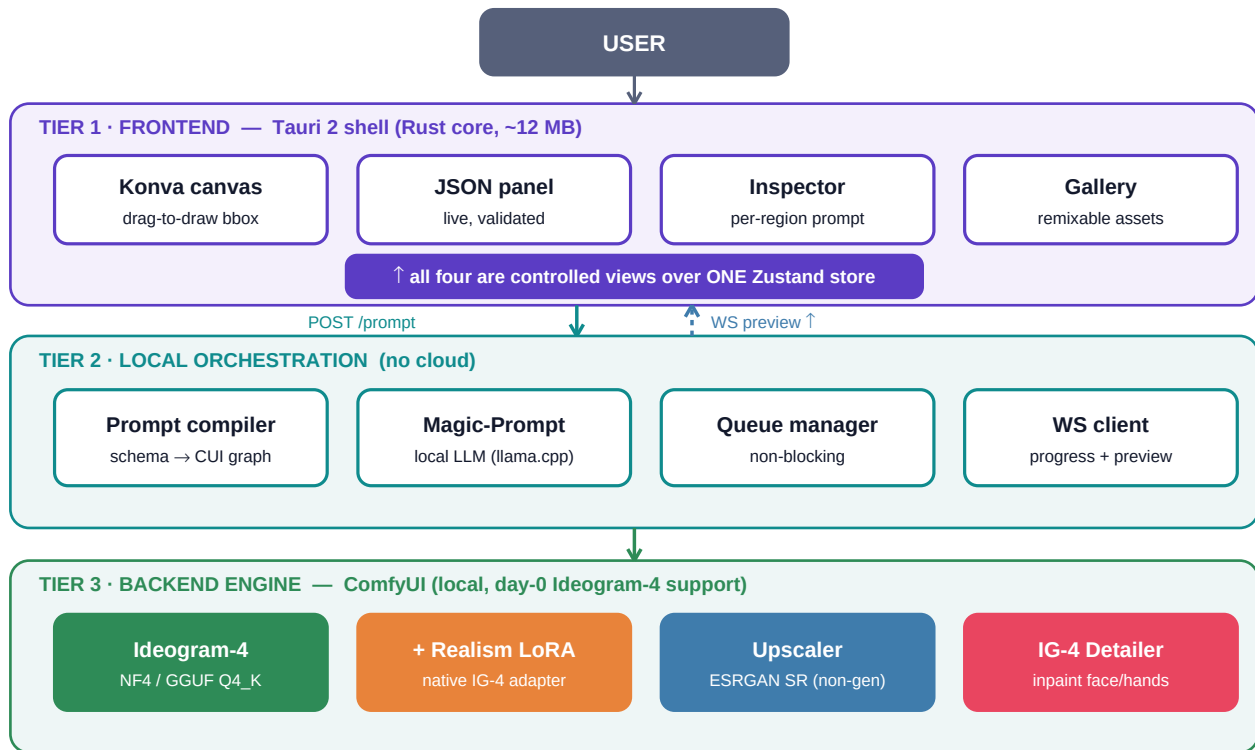


Fig. 1 — Three tiers: a Tauri/React frontend with a Konva bbox canvas, a local orchestration layer that compiles the structured prompt into a ComfyUI graph, and ComfyUI running Ideogram-4 + realism add-ons.

Stack rationale

- **Shell** — **Tauri 2** over Electron: ~12 MB vs ~180 MB, 30–50 MB idle RAM, faster start, sandboxed FS access — it matters next to a GPU-hungry backend. (Electron is the fallback if you want zero WebView quirks.)
- **Frontend** — **React + TypeScript + Konva.js** (the stack InvokeAI ships): official React bindings, a built-in Transformer for resize handles, multi-layer canvas, snapping/bounded drag.
- **State** — **Zustand**: one normalized store is the single source of truth for the canvas, inspector, and JSON views.
- **Backend** — **ComfyUI** via `POST /prompt` + a `ws://` connection for live progress and latent previews.

5 · The killer feature — a visual bbox <-> JSON editor

This is the heart of the product and the direct antidote to pain points 1–4. One store, three synchronized views, with two-way binding so the canvas and the JSON never drift.

controlled views — last valid edit wins; invalid JSON keeps last good boxes

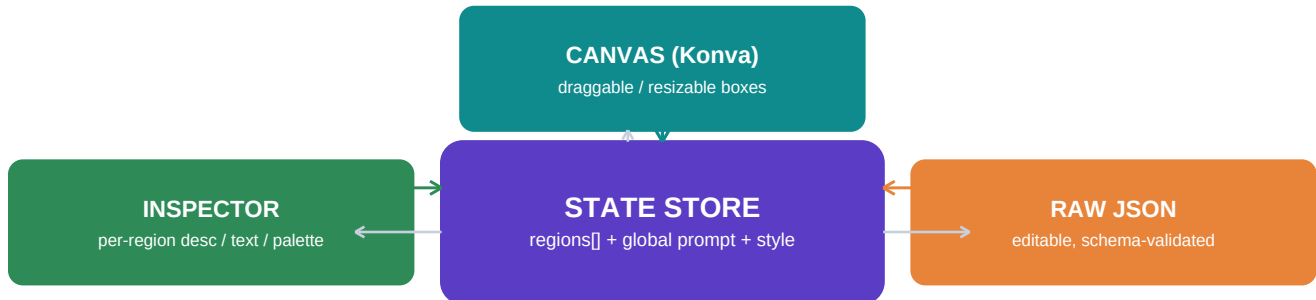


Fig. 2 — Two-way sync. Dragging a box normalizes to 0–1000 and writes the store; editing JSON parses/validates and updates boxes. The store is authoritative.

The structured-prompt model it edits (Ideogram-4 schema)

```
{
  "high_level_description": "...",
  "style_description": {
    "photo" | "art_style": "...", // exactly one
    "aesthetics": "...", "lighting": "...", "medium": "...",
    "color_palette": ["#RRGGBB", ...] // <= 16, UPPERCASE
  },
  "compositional_deconstruction": {
    "background": "...",
    "elements": [
      { "type": "text", "bbox": [y_min, x_min, y_max, x_max], // 0-1000, y-first
        "text": "GRAND OPENING", "desc": "bold serif, gold" },
      { "type": "obj", "bbox": [450, 550, 880, 950], "desc": "glass juice bottle" }
    ]
  }
}
```

Implementation notes: serialize with fixed key order and compact separators (the model is order-sensitive); enforce schema live (one of photo/art_style, uppercase hex, <=16 colors); warn when a box covers <2% of canvas (spatial conditioning is low-res). The [benjiyaya/ComfyUI-Ideogram4-Toolkit](#) already provides a drag-to-draw 1000×1000 canvas node you can learn from or wrap.

6 · Running Ideogram 4 on the RTX 4090

Ideogram 4 is a 9.3B single-stream diffusion transformer (34 layers, flow-matching) with a frozen **Qwen3-VL-8B** text encoder and an 8× VAE, native 256–2048 px. The text encoder is the VRAM pressure point — and the key to fitting 24 GB.

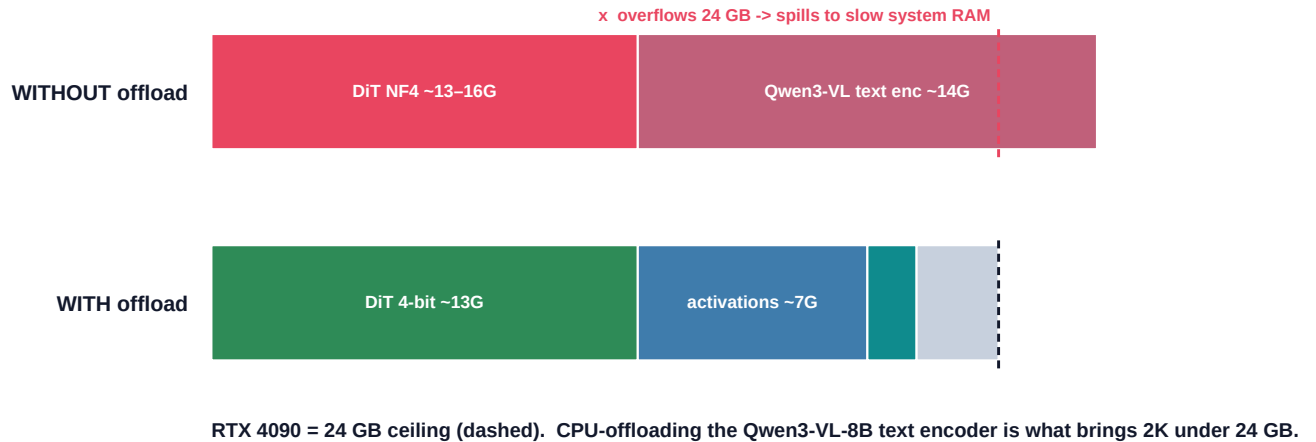


Fig. 3 — VRAM budget. Resident, the DiT + text encoder overflow 24 GB; CPU-offloading the Qwen3-VL encoder brings a 2K generation under the ceiling.

Spec	Value
Params / arch	9.3B DiT, 34 layers, flow-matching, Euler ODE sampler
Text encoder	Qwen3-VL-8B-Instruct (frozen; ~14 GB if resident → offload it)
Checkpoints	ideogram-4-nf4, -fp8; Comfy-Org mirror; GGUF Q4_K community builds
Fits 24 GB?	NF4 / Q4_K yes (tight, batch 1 at 2K). FP8 needs 32 GB+ — not on a 4090.
Speed (4090, 4-bit)	~21 s 1K turbo · ~32 s default · ~72 s quality · ~60 s at 2K (directional)
Layout interface	Structured JSON is the native conditioning — no ControlNet needed for bbox

CHECKPOINT PICK

Start on **NF4** (guaranteed Diffusers/ComfyUI support). If text fidelity matters — and it's Ideogram's whole point — switch to a **GGUF Q4_K** build via city96/ComfyUI-GGUF: research reports it beats NF4 quality at the same size, and NF4 specifically degrades lettering. Benchmark both on your card.

7 · The hyper-realism pipeline — Ideogram 4 only

Per your constraint, **every generative step is Ideogram 4** — no Flux, no SDXL, no second model. Realism comes from three IG-4-native levers (a native realism LoRA, photo-style JSON prompt craft, and IG-4's own img2img/inpaint passes) plus one non-generative helper (a plain super-resolution upscaler that only adds pixels, not style).

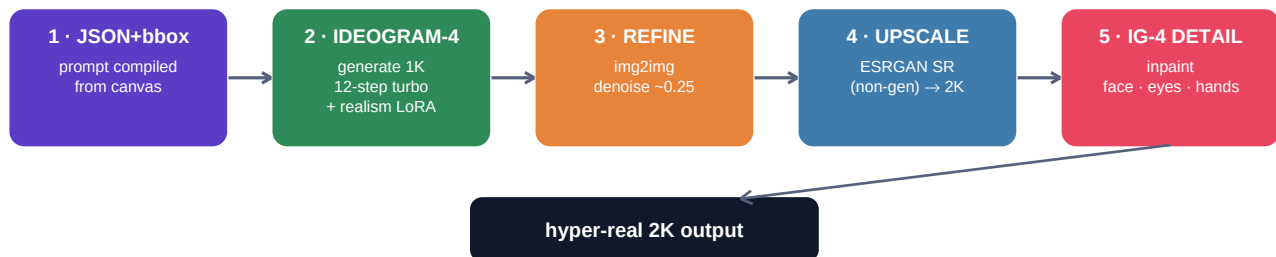


Fig. 4 — A single-model chain: IG-4 generates with a native realism LoRA, IG-4 refines at low denoise, a non-generative upscaler enlarges to 2K, then IG-4 re-renders face/hands by inpaint. (Time grows super-linearly with pixels, so small→upscale beats native 2K.)

STAYING SINGLE-MODEL (by design)

The only non-IG-4 pieces are a **super-resolution upscaler** (e.g. an ESRGAN net — pure math, generates no new content/style) and a tiny **face/hand detector** that merely *selects* the region Ideogram 4 then re-renders. The popular Civitai realism LoRAs are trained on Flux / SDXL / Pony / Z-Image and **cannot load on Ideogram 4's transformer** — they are intentionally excluded. If you want a fully pure run, replace the upscaler with IG-4 **tiled img2img** upscaling so 100% of pixels come from Ideogram 4.

8 · Making Ideogram 4 *itself* photoreal

With one model, realism is won on three IG-4-native fronts. The biggest lever you fully control — and it costs nothing extra — is **prompt craft**.

A · Photo-style JSON prompt craft (zero extra models)

- Use `style_description.photo` (never `art_style`) and describe a real camera: *“shot on 85mm f/1.4, shallow depth of field, Kodak Portra grain.”*
- Specify **physical lighting** (softbox, golden hour, window light) and **skin/texture** words: pores, fine lines, stray hairs, subsurface scattering.
- Add **imperfection** cues — candid framing, slight motion, asymmetry — and **avoid** “perfect / flawless / hyper-detailed,” which push the plastic AI look.

B · A native Ideogram-4 realism LoRA (the only kind that loads)

- Source from the early IG-4 LoRA collections (e.g. DeverStyle/Ideogram-4.0-Loras on HuggingFace) **or train your own** — the reliable path to a specific look.
- **Training:** use **ai-toolkit** (kohya_ss doesn't support IG-4 yet) or the hosted **fal.ai V4 trainer**; rank 16–64, 20–100 images, **captioned in the structured-JSON format**. Feasible on ~12 GB, so your 4090 trains comfortably. Apply at ~0.6 strength.

ON YOUR “REALISM v5”

It won't load unless it is specifically an **Ideogram-4** LoRA. RealVisXL V5.0 (SDXL), the Z-Image “Realistic Snapshot v5,” and the Flux realism LoRAs are all incompatible with IG-4. To get that exact look on a single-model IG-4 setup, **train an IG-4 realism LoRA on a dataset in that style** (or find an IG-4-native port). Send me the link when you have it and I'll confirm.

C · IG-4 refine & detail passes (within the same model)

- **Refine:** a low-denoise (~0.2–0.35) IG-4 `img2img` pass over the first render adds micro-texture without changing composition.
- **Detail:** detect the face/eyes/hands, then have **IG-4 inpaint** just that crop at higher detail — the single biggest portrait-realism win.
- **Caveat:** IG-4's open pipeline `img2img/inpaint` exposure is only *partly* confirmed locally (editable/inpaint features were described as a follow-up release). If local IG-4 inpaint is limited today, these passes wait on that capability — we do **not** substitute another model.

D · Upscaling (non-generative)

A plain ESRGAN super-resolution model (e.g. 4x-UltraSharp) only raises resolution — it adds no model style, so it stays true to a single-model philosophy. Use tiled decode to fit 24 GB. For a 100%-IG-4 run, use IG-4 tiled `img2img` upscaling instead.

9 · 4090 performance tuning checklist

Lever	Action	Payoff
Quantization	4-bit NF4 (compat) or GGUF Q4_K (text); never FP8 on 24 GB	Fits the card
Text-encoder offload	CPU-offload Qwen3-VL-8B (~14 GB)	Brings 2K under ceiling
Tiled VAE decode	Tile the 2K decode stage	Avoids decode spike
Attention	Flash / SageAttention (8-bit attn)	'Significant' speedup
Step caching	TeaCache-style; Sage+TeaCache $\approx$ ~2 min, Flash+TeaCache $\approx$ ~1 min/img	Fewer effective steps
torch.compile	Enable	Modest (~6%)
Sampler presets	12-step turbo for previews; 48-step for finals	Draft fast, finalize sharp
Resolution strategy	Generate 1K $\rightarrow$ upscale ($4\times$ area $\approx 6.3\times$ time)	Beats native 2K
Batch / hygiene	Batch 1 at 2K; restart backend to clear VRAM before big runs	Stability

10 · 'Premium feel' & build roadmap

Premium-feel feature set

- **Live latent previews** streamed over the ComfyUI WebSocket (TAESD), throttled ~2–4 fps; per-step progress on the canvas.
- **Non-blocking queue** — enqueue many jobs, keep editing; cancel/reorder.
- **Two-way undo/redo** over the store (canvas *and* JSON), keyboard shortcuts for everything, nudge boxes with arrow keys.
- **Gallery with full provenance** — store the exact JSON + seed + graph with each image; 'remix' loads it back into the editor; drag a result onto a region as a reference.
- **Magic-Prompt (local)** — a local LLM expands a one-line idea into schema-valid JSON with sensible boxes.
- **Dark-first UI** — 4+ surface levels, separation by luminance/borders not shadows (Linear/Raycast feel), smooth GPU canvas pan/zoom, subtle micro-interactions.

Phased roadmap

Phase	Scope	Outcome
0 · Spike	ComfyUI + Ideogram-4 NF4 on the 4090; confirm speeds + offload	De-risk the engine
1 · MVP	Tauri+React shell, Konva bbox canvas, JSON two-way sync, single generate	The core 'aha' editor
2 · Realism	Refine→upscale→detail graph; IG-4 LoRA loader; preset library	Hyper-real output
3 · Premium	Live previews, queue, gallery/provenance, undo, shortcuts, dark UI	'Premium feel'
4 · Edit	Mask/inpaint region edits + seed-lock to approximate localized edits	Smooths pain #4

BOTTOM LINE

The fastest credible build is **ComfyUI (engine) + a Tauri/React/Konva front-end** whose star is the **visual bbox-to-JSON editor**. That single feature neutralizes Ideogram 4's four loudest complaints, and self-hosting removes credits, queues, and the censorship gate. Mind the non-commercial license and IG-4's photoreal ceiling — push realism with a native IG-4 LoRA, photo-style JSON prompt craft, and IG-4's own refine/inpaint passes; no second model.

Appendix · Key sources

Model & local engine

- HuggingFace: [ideogram-ai/ideogram-4-nf4](#) · [-fp8](#) · [Comfy-Org/Ideogram-4](#) · [transformerlab/ideogram-4-gguf-q4_k](#)
- GitHub: [ideogram-oss/ideogram4](#) ([docs/pipeline.md](#), [docs/prompting.md](#)) · [ideogram-oss/ComfyUI-Ideogram4](#) · [benjiyaya/ComfyUI-Ideogram4-Toolkit](#) · [city96/ComfyUI-GGUF](#)
- ComfyUI day-0 support & tutorial ([comfyui.org](#), [docs.comfy.org](#)) · Diffusers Ideogram4 pipeline docs · Transformer Lab quantization study ([lab.cloud](#)) · [omkamal '4090 structured JSON' benchmark](#).

Pain points & UX

- Ideogram 4 reviews ([goenhance 'messy open-weight story'](#), [seaart](#)) · [safety-filter GitHub issue](#) · [ideogram-oss/ideogram4#5](#) · [pricing/queue threads](#) ([eesel](#), [Aiville](#), [Trustpilot](#)) · [Ideogram on X re: queue scheduling](#).
- Local UI landscape: [ComfyUI](#) / [InvokeAI](#) ([Konva canvas](#), [regional guidance](#)) / [Krita AI Diffusion](#) / [SwarmUI](#) / [Fooocus](#) / [Forge](#). [Tauri-vs-Electron](#) & [Konva-vs-Fabric](#) write-ups.

Realism (Ideogram-4 native)

- IG-4 LoRA: hosted [fal.ai V4 trainer](#) · [community collection](#) [DeverStyle/Ideogram-4.0-Loras](#) (HF) · [ai-toolkit](#) local training guide (~12 GB).
- IG-4 prompt craft & structured-JSON captioning: [ideogram-oss/ideogram4 docs/prompting.md](#) · [ImagineArt JSON prompt guide](#) · ['fix plastic skin in ComfyUI' write-up](#). [ESRGAN super-resolution](#) (e.g. [4x-UltraSharp](#)) as a non-generative upscaler.

RELIABILITY NOTE

Ideogram 4 released after the assistant's training cutoff, so all model facts come from June 2026 web research. Speed figures and the 'GGUF beats NF4' quality claim lean on one or two detailed sources — treat as directional and benchmark on your hardware. Several sites (Civitai, some blogs) block direct fetch; their URLs are real but verify specifics on-page. The non-commercial license is well-corroborated and is the most important item to confirm firsthand.